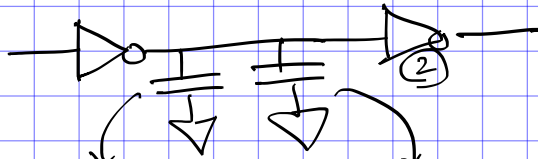


Logical effort



$C_{intrinsic}$ or C_{self}

$C_{Load} = \text{gate cap of (2)} \rightarrow \text{also called } C_{inv}$

Propagation delay $t_p = \frac{t_{PHL} + t_{PLH}}{2}$

$$= \underbrace{\left(R_{eqp} + R_{eqn} \right)}_{R_{inv}} C_{total} \times \ln(2)$$

$$= 0.69 R_{inv} (C_{Load} + C_{self})$$

$$= 0.69 R_{inv} C_{inv} \left(\frac{C_{Load}}{C_{inv}} + \frac{C_{self}}{C_{inv}} \right)$$

gate cap of unit inverter $= 3 C_g$

$B \rightarrow \text{approx. 1}$

gate cap of unit MOS $\left(\frac{0.12}{0.15} \right) = \left(\frac{W}{L} \right)_n$

$$t_p = 0.69 R_{inv} C_{inv} \left(\frac{C_{Load}}{C_{in}} \times \frac{C_{in}}{C_{inv}} + \frac{C_{self}}{C_{inv}} \right)$$

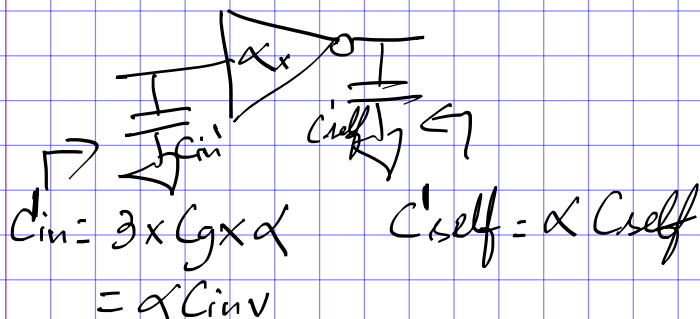
Electrical effort (n) Logical eff. (g) parasitic (p)

parasitic delay or inertial delay.

$$t_p = \tau(p + g h) = \tau(\beta + h)$$

$\tau = R_{inv} C_{inv} \ln(2)$
 $\quad \quad \quad = 1 \text{ for inverter}$

Consider a scaled inverter with PMOS widths scaled by a factor (α).



$$R_{eqn} = \frac{R_{eqn}}{\alpha}$$

$$R_{eqp} = \frac{R_{eqp}}{\alpha}$$

$$\Rightarrow R_{inv} = R_{inv} / \alpha$$

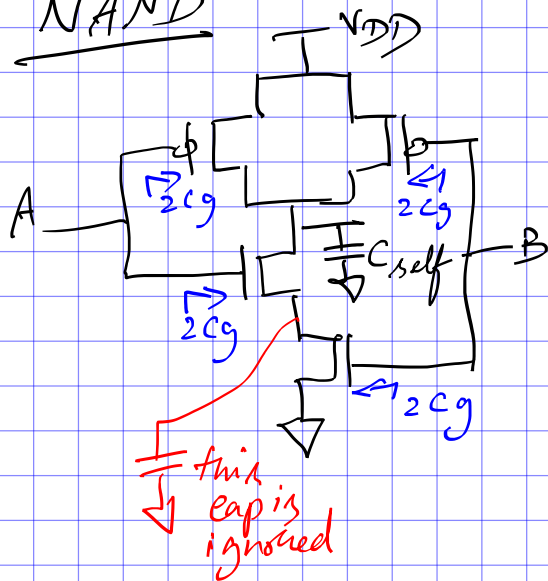
Delay of this inverter $d = 0.69 R_{inv} C_{inv} \left(\frac{C_{load}}{C_{in}} \times \frac{C_{in}}{C_{inv}} + \frac{C_{self}}{C_{inv}} \right)$

$$= 0.69 \frac{R_{inv}}{\alpha} C_{inv} \left(\frac{C_{load}}{C_{in}} \times \cancel{\frac{C_{in}}{C_{inv}}} + \frac{C_{self}}{C_{inv}} \right)$$

$$= \tau(h + \beta)$$

We get an expression to specify the delay of arbitrarily scaled versions similar to the unit inv.

NAND



$$C_{self} = 6 \times C_g \times \beta$$

$$C_{in} = 4C_g \text{ for each input.}$$

$$= \frac{4}{3} C_{inv}$$

$$d_{NAND} = 0.69 R_{inv} (C_{load} + C_{self})$$

$$= 0.69 R_{inv} C_{inv} \left(\frac{C_{load}}{C_{in}} \times \frac{C_{in}}{C_{inv}} + \frac{C_{self}}{C_{inv}} \right)$$

$$= \tau \left(h \times \frac{4}{3} + 2\beta \right)$$

Similarly for NOR gate $d_{NOR} = \tau \left(h \times \frac{5}{3} + 2\beta \right)$

HW find the delay expressions for N-input NAND/NOR

An alternate way to calculate logical effort and parasitic delay of static CMOS is to ensure first that the worst case resistance matches that of the unit inverter for both PUN & PDN.

$$g = \frac{\text{input cap.}}{1 \times \text{inv. input cap.}}$$

$$p = \frac{\text{self cap at output}}{C_{self} \text{ of } 1 \times \text{inverter}} \times \beta$$

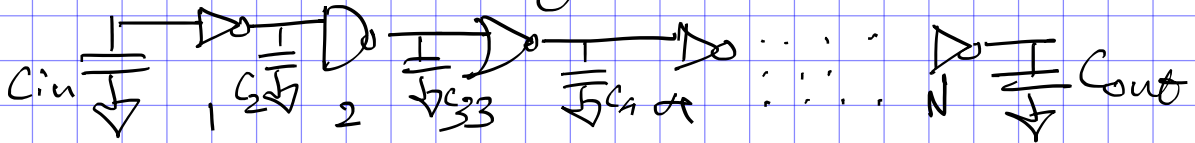
e.g. for 2-input NAND

$$C_{\text{input}} \text{ for each input} = 4C_g$$

$$C_{\text{self}} = 6C_g$$

$$g = \frac{4C_g}{3C_g}, \quad p = \frac{6C_g}{3C_g} = 2$$

Using the same methodology let's calculate the delay of paths containing multiple gates.



Delay of i^{th} gate, $d_i = g_i h_i + p_i$

$$h_i = \frac{C_{i+1}}{C_i} \left\{ \begin{array}{l} \text{where } C_i \text{ is the input cap} \\ \text{of the } i^{\text{th}} \text{ gate.} \end{array} \right.$$

$$h_1 \times h_2 \times h_3 \cdots h_N = \frac{C_2}{C_1} \times \frac{C_3}{C_2} \times \frac{C_4}{C_3} \cdots \frac{C_{out}}{C_N}$$

$$= \frac{C_{out}}{C_{in}} = H \left\{ \begin{array}{l} \text{electrical} \\ \text{effort of the} \\ \text{path} \end{array} \right.$$

$$h_N = \frac{H}{\prod_{i=1}^{N-1} h_i}$$

Delay of the path is the sum of the delays of the constituent gates

$$D = \sum_{i=1}^N d_i = \sum_{i=1}^N (p_i + g_i h_i)$$

$$= \underbrace{\sum_{i=1}^N p_i}_P + \sum_{i=1}^N g_i h_i$$

$P \rightarrow$ parasitic delay of path

$$\sum_{i=1}^N g_i h_i = \sum_{i=1}^{N-1} g_i h_i + g_N \times \frac{H}{\prod_{i=1}^{N-1} h_i} = D - P$$

To minimise delay D ,

$$\frac{\partial D}{\partial h_1} = g_1 + g_N \times \frac{H}{\prod_{i=1}^{N-1} h_i} \times \left(\frac{-1}{h_1} \right) = 0$$

$$\Rightarrow g_1 h_1 = g_N h_N$$

This holds good for all h_i

$$\Rightarrow g_i h_i = g_N h_N, \forall i \in \{1, 2, \dots, N\}$$

$$\prod_{i=1}^N g_i h_i = \left(\prod_{i=1}^N g_i \right) H = (g_i h_i)^N$$

G_1 : path logical effort

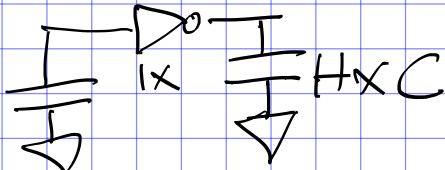
$$g_i h_i = (G_1 H)^{1/N}$$

g_i : stage effort

$$D = P + N \times (G_1 H)^{1/N}$$

Special case of a path of inverters driving a large load. The goal is to find the number of inverters that minimizes the delay.

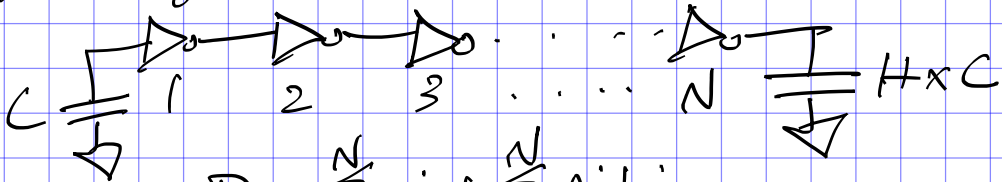
Delay of a single stage



$$D = p + gH = 1 + H$$

$\frac{1}{3}Cg$

Delay of N stages:



$$D = \sum_{i=1}^N p_i + \sum_{i=1}^N g_i t_i$$

min delay $D = N + H^{\frac{1}{N}} \times N$ since $\sum_{i=1}^N \frac{1}{g_i} = 1$

to get optimal number of inverters

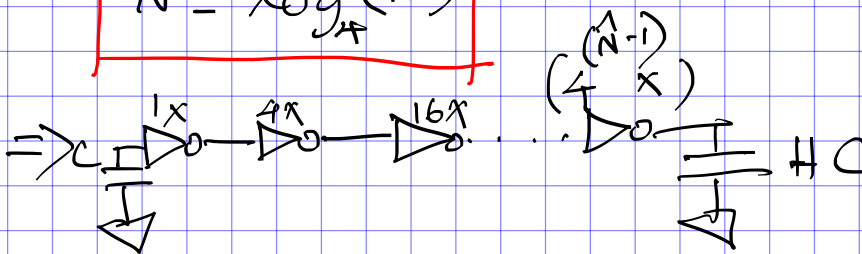
$$\frac{\partial D}{\partial N} = 0 = 1 + H^{\frac{1}{N}} - H^{\frac{1}{N}} \ln(H^{\frac{1}{N}})$$

$$\Rightarrow H^{\frac{1}{N}} \approx 3.59$$

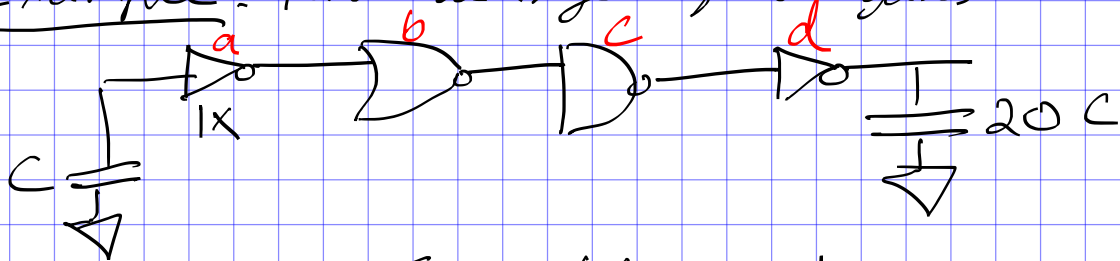
$$\hat{N} = \log_4(H)$$

$$g_i \approx 3.59 \text{ snapped to } 4$$

FOH delay
Fanout 4



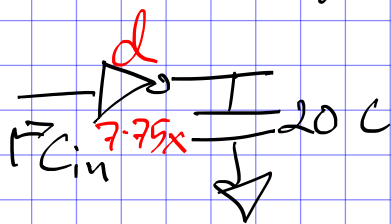
Example: Find the sizes of all gates that min. D.



g	1	$\frac{5}{3}$	$\frac{4}{3}$	1
p	1	2	2	1

$$P = \sum_{i=1}^4 p_i = 6, \quad G = \prod_{i=1}^4 p_i = \frac{20}{9}, \quad H = \frac{20C}{C} = 20$$

$$f_i^* = g; h_i = (GH)^{\frac{1}{4}} \approx 2.58$$

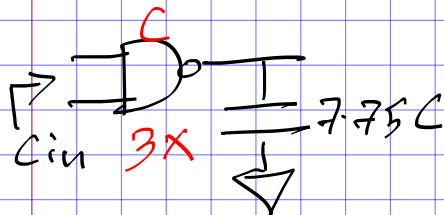


$$g; h_i = 2.58$$

$$1 \times \frac{20C}{C_{in}} = 2.58 \Rightarrow C_{in} \approx 7.75C$$

$$\Rightarrow C_{in} \approx 7.75C$$

$$\text{size} = \frac{7.75C}{C} = 7.75$$



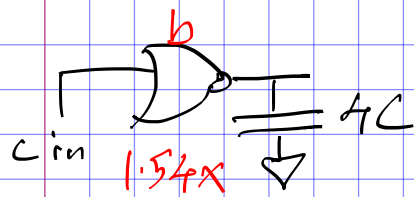
$$\frac{4}{3} \times \frac{7.75C}{C_{in}} = 2.58$$

$$\Rightarrow C_{in} \approx 4C$$

$$\text{size} = \frac{4 \times C}{(\frac{4}{3}C)}$$

$$= 3$$

4Cg is the looking in cap of 1x NAND

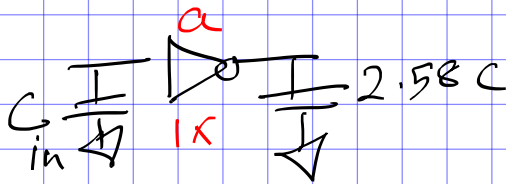


$$\frac{5}{3} \times \frac{4C}{C_{in}} = 2.58 \Rightarrow C_{in} = 2.58C$$

size of NOR

$$= \frac{2.58C}{\frac{5}{3}C} \approx 1.54x$$

Check that the inverter has a size of 1x as assumed.

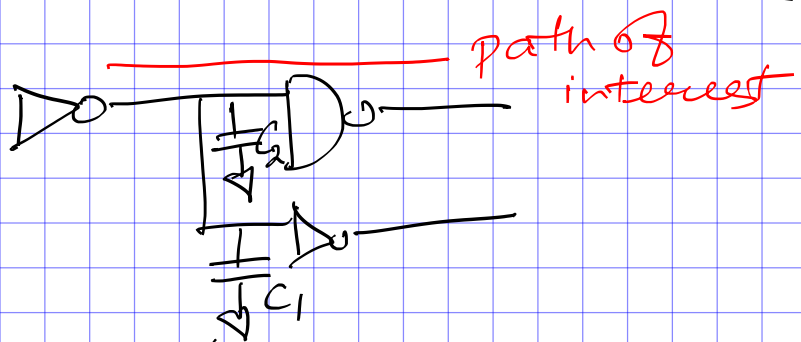


$$1 \times \frac{2.58C}{C_{in}} = \underbrace{2.58}_{f_i^*}$$

$C_{in} = C$

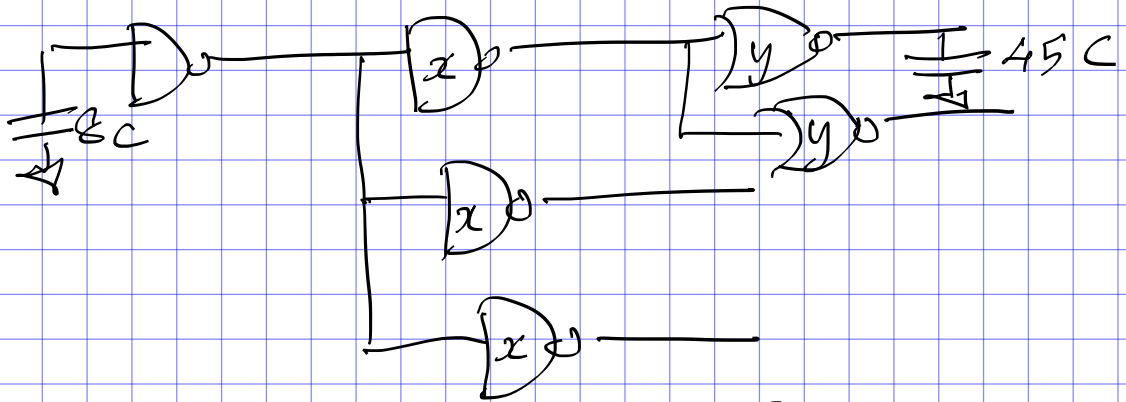
size of inv = $\frac{C}{C} = 1x$

Branching



Branching effort $b_i = \frac{C_{onpath} + C_{offpath}}{C_{onpath}}$

$$= \frac{C_1 + C_2}{C_2}$$



g	$\frac{4}{3}$	$\frac{4}{3}$	$\frac{5}{3}$
p	2	2	2
b	$\frac{x+2x}{x} = 3$	$\frac{y+y}{y} = 2$	1

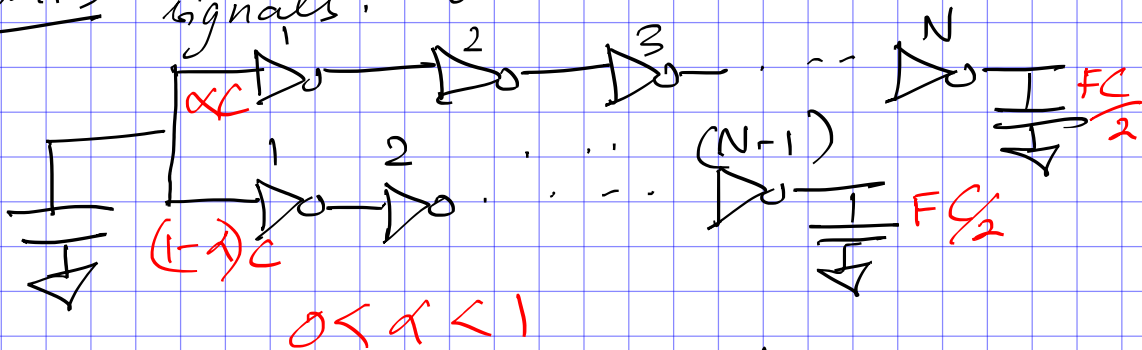
$$F = G_1 B H = \pi g_i \pi b_i \pi h_i$$

$$= \frac{80}{27} \times 6 \times \frac{45}{8} = 100$$

$$f_i^* = g_i h_i b_i = F^{\frac{1}{3}} = (100)^{\frac{1}{3}} \approx 4.64$$

HW calculate sizes 'x' & 'y'

Forks: Match delays of true & complementary signals.

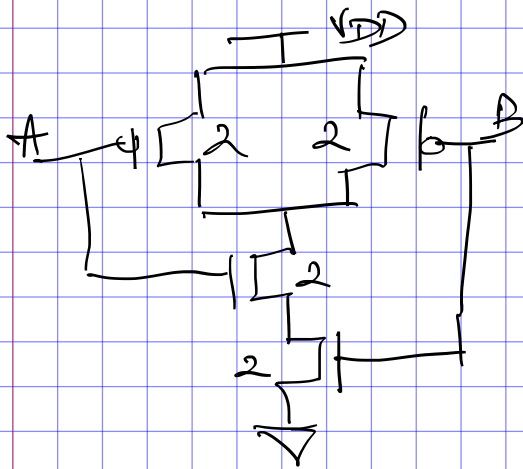


$$\underbrace{\left(\frac{F}{2x}\right)^{\frac{1}{N}} \times N + N}_{\text{min. delay of top path}} = \underbrace{\left(\frac{F}{2(1-x)}\right)^{\frac{1}{N-1}} \times (N-1) + (N-1)}_{\text{min. delay of bottom path.}}$$

Fork to use for various ranges of F

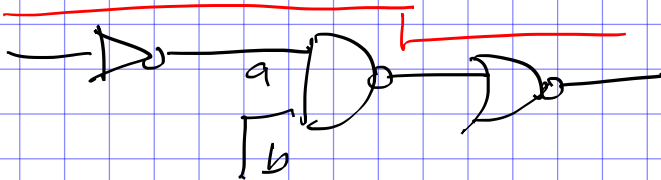
Fork	$F_{\min}$	$F_{\max}$
2-1	2	9.6
3-2	9.6	38.7
4-3	38.7	146
5-4	146	538
6-5	538	1970
7-6	1970	7150

Symmetric and Asymmetric gates



input	A	B
g	$\frac{4}{3}$	$\frac{4}{3}$
p	2	2

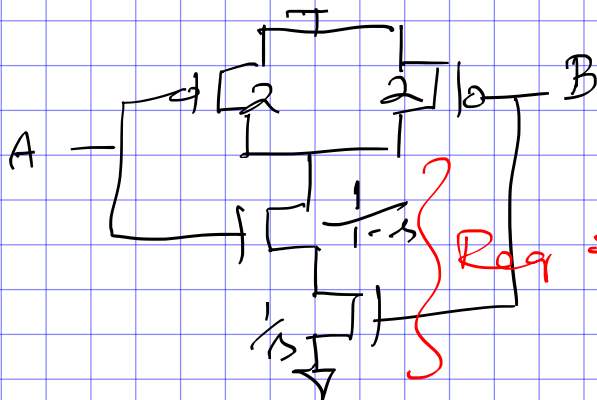
Suppose one of the inputs is in one path of choice to minimise the delay.



We can minimise the delay through one of the inputs by increasing the delay through other inputs.

s : symmetry factor $0 < s < 1$

$s = \frac{1}{2}$ vanilla gate with both inputs symm.



$$R_{eq} = \frac{R_{eqn}}{\frac{1}{1-s}} + \frac{R_{eqn}}{\frac{1}{s}}$$

$$= R_{eqn}$$

input	A	B	
g	$\frac{3-2s}{3-3s}$	$\frac{2s+1}{3s}$	$s = \frac{1}{2}$ $\frac{4}{3}$
p	$\frac{5-4s}{3-3s}$	$\frac{5-4s}{3-3s}$	2

Let us make $s \approx 0.01 \Rightarrow g_A \approx 1, g_B \approx 34$

$$p \approx \frac{5}{3}$$

$$d_A = \frac{5}{3} + h$$

$$d_B = \frac{5}{3} + 34h$$

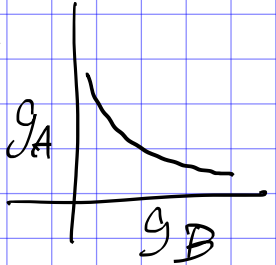
Vanilla
NAND

$$d = 2 + \frac{4}{3}h$$

$$d_A < d < d_B$$

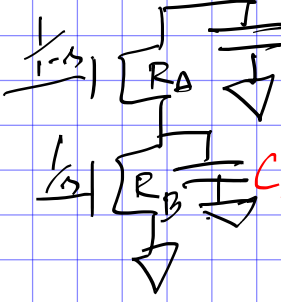
for range h.

$$g_A - 1 = \frac{s}{3-3s}, \quad g_B - 1 = \frac{1-s}{3s}$$

$$(g_A - 1)(g_B - 1) = \frac{1}{9}$$


's' should be $< \frac{1}{2} \Rightarrow$ size of A $<$ size of B

Reason:



$$R_A = R_{eqn} \times (1-s)$$

$$R_B = R_{eqn} \times s$$

$$C_{interim} = \frac{C_{int}}{1-s} + \frac{C_{int}}{s} = \frac{C_{int}}{s(1-s)}$$

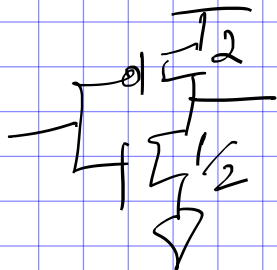
C_{int} = depletion cap of unit MOS

Elmore delay of this network = $(R_A + R_B)C_{Total}$
 $+ R_B \times C_{interim}$
 $= R_{eqn}C_{Total} + R_B \times C_{interim}$

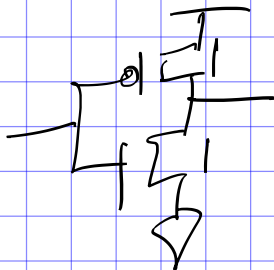
$C_{interim}$ is a constant + irrespective of which among A or B is larger. To minimize the delay, R_B has to be smaller than $R_A \Rightarrow$ B has to be larger than A $\Rightarrow s < \frac{1}{2}$.

Skewed gates

Hi skew

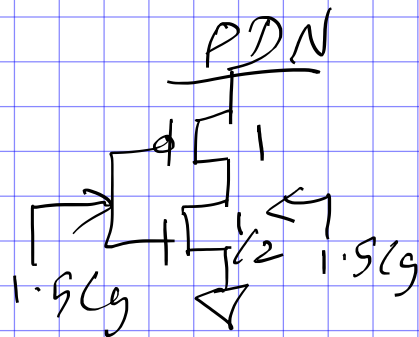
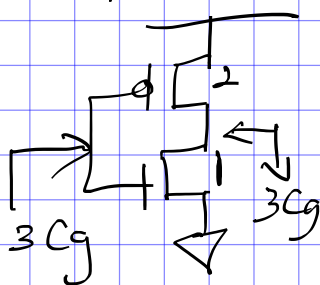
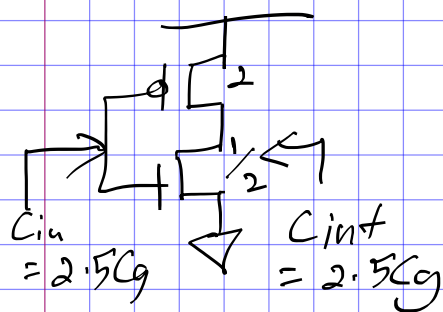


Lo skew



Calculation of logical/parasitic effort for Hi skew

Equivalent inverter PUN



g

$$\frac{5}{6} < 1$$

$$\frac{5}{3} > 1$$

P

$$\frac{5}{6}$$

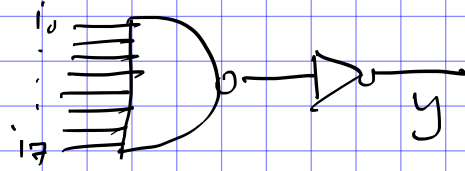
$$\frac{5}{3}$$

reference inverter $d = 1 + h$

better pullup delay $= \frac{5}{6} + \frac{5}{6}h$

worse pull down $= \frac{5}{3} + \frac{5}{3}h$

Multiple implementations of 8-input AND gate

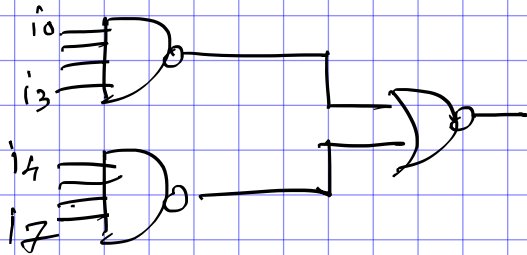


$$\begin{array}{ccc}
 g & 10/3 & 1 \\
 P & 8 & 1
 \end{array}$$

$$G_1 = 10/3 \quad H$$

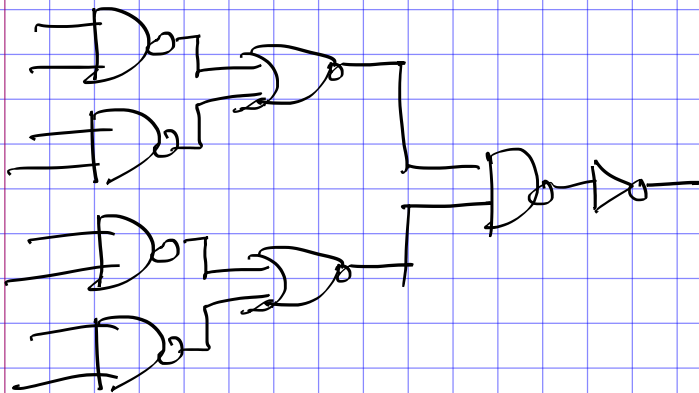
$$f_1^* = \sqrt{\frac{10H}{3}}$$

$$D = \left(9 + 2 \sqrt{\frac{10H}{3}} \right) t_{pinv}$$



$$\begin{array}{ccc}
 g & 2 & 5/3 \\
 P & 4 & 2
 \end{array}$$

$$D = \left(6 + 2 \sqrt{\frac{10H}{3}} \right) t_{pinv}$$

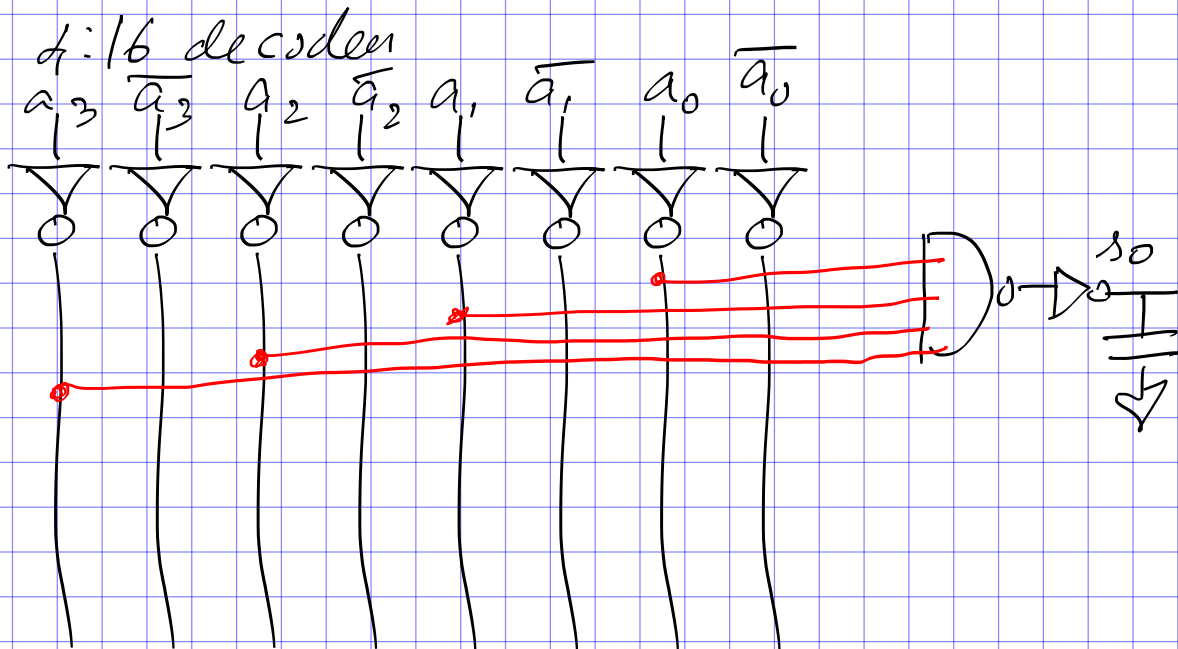


$$\begin{array}{ccccc}
 P & 2 & 2 & 2 & 1 \\
 g & 4/3 & 5/3 & 7/3 & 1
 \end{array}$$

$$D = \left[7 + 4 \left(\frac{80H}{27} \right)^{1/4} \right] t_{pinv}$$

+1W Calculate delays
for $H = 2, 10, 20$
and see which version
has minimum delay.

Decoders in a register file



$$s_0 = \bar{a}_3 \bar{a}_2 \bar{a}_1 \bar{a}_0$$

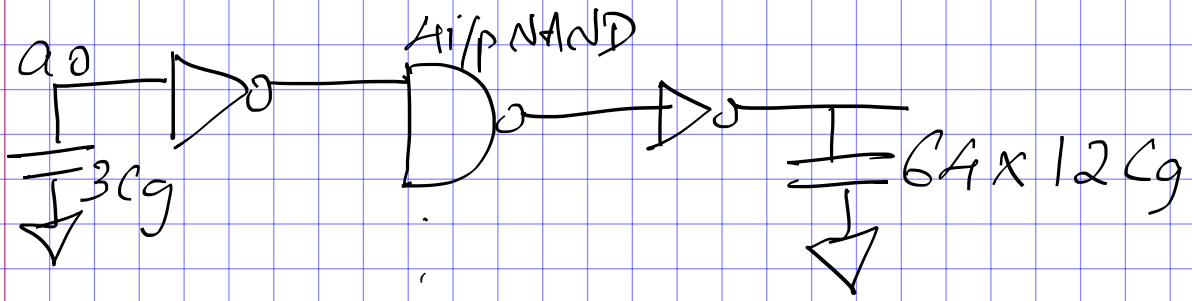
$$s_1 = \bar{a}_3 \bar{a}_2 \bar{a}_1 a_0$$

⋮

$$s_7 = a_3 a_2 a_1 a_0$$

Each decoder
drives 64 tristate
inverters.

All paths are
symmetric & have
to be simultaneously
optimised.



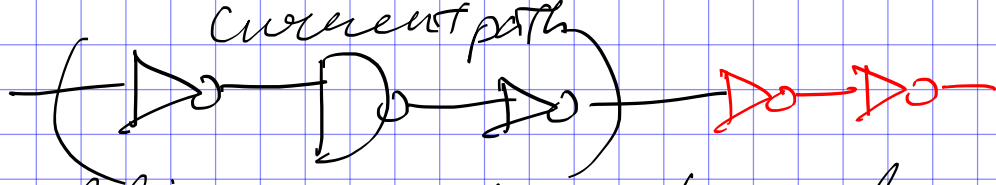
8 copies $\left\{ \begin{array}{l} a_0 \text{ drives 8 copies} \\ \bar{a}_0 \text{ drives 8 copies} \end{array} \right.$

P	1	4	1	$F = GBH$ $= 2 \times 8 \times (64 \times 4)$ $f_i^* = F^{1/3} = 16$
g	1	2	1	
b	1	8	1	

HW: calculate sizes of the gates.

Optimal stage effort $f_i^* = 4$

$\Rightarrow$ the path can be appended with more inverters to reduce stage effort.

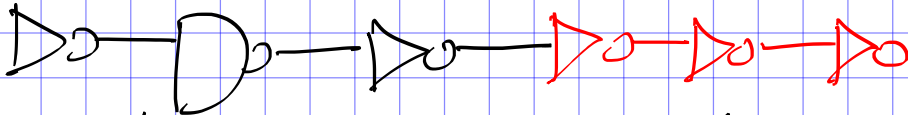


Adding more inverters doesn't affect F .

Optimal number of stages

$$N^* = \log_4 F = \log_4 (64^2) = 6$$

$\Rightarrow$



achieves the optimal stage effort

$$D = P + N^* (GBH)^N$$

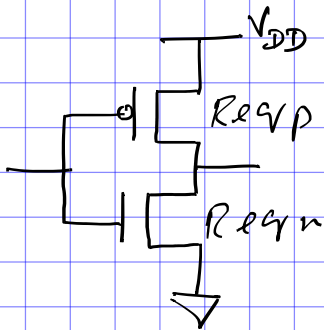
$$= P + N^* \times 4$$

HW Find the optimal delay and the size of all gates in the path.

Sizing the PMOS in reference inverters to minimise delay.

* 2:1 size of PMOS was obtained by making

$V_{th} = V_{DD}/2$ which ensures symmetric NM_H & NM_L .
Ignoring channel length modulation



$$R_{eqP} \propto \frac{C_{ox} \text{ (Gate) } V_{DD}}{I_{DSATP}} \propto \frac{1}{\mu_p W_p}$$

$$\text{III}^{\text{ly}} R_{eqN} \propto \frac{1}{\mu_n W_n}$$

$$C_{eff} \propto (W_n + W_p) \quad \left| \begin{array}{l} \text{Ignoring} \\ \text{the sidewall} \\ \text{Leff dep.} \end{array} \right.$$

$$t_{pinv} \propto R_{eq} \times C_{eff} = \frac{(R_{eqP} + R_{eqN})}{2} C_{eff}$$

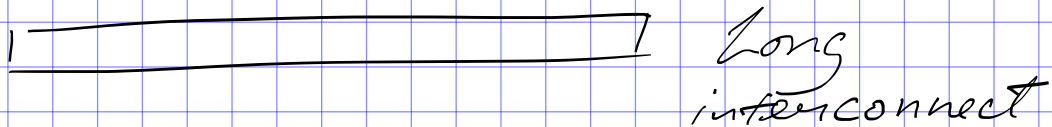
$$\propto \left(\frac{1}{\mu_n W_n} + \frac{1}{\mu_p W_p} \right) \times (W_p + W_n)$$

To minimise delay

$$0 = \frac{dt_{pinv}}{dW_p} = -\frac{1}{\mu_p W_p^2} (W_p + W_n) + \frac{1}{\mu_n W_n} + \frac{1}{\mu_p W_p}$$

$$\Rightarrow \frac{W_n}{\mu_p W_p^2} = \frac{1}{\mu_n W_n} \Rightarrow \boxed{W_p = \sqrt{\frac{\mu_p}{\mu_n}} \times W_n}$$

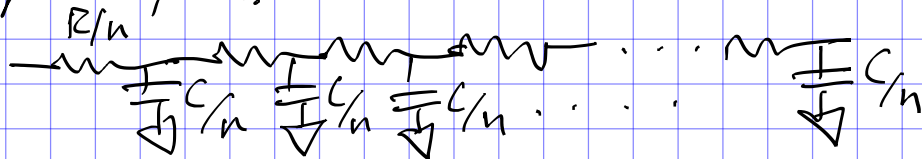
Interconnect Delay Optimization By Buffering



Multiple ways of modeling

1.) Single lumped R, C $\tau = RC \ln(2)$

2.) Multiple R, C sections.



$$r = R/n, \quad \kappa = C/n$$

→ Elmore

$$\begin{aligned} \tau &= (nr\kappa + (n-1)r\kappa + \dots + r\kappa) \ln 2 \\ &= r\kappa \frac{n(n+1)}{2} \end{aligned}$$

$$= (R/n)(C/n) \left(1 + \frac{1}{n}\right) \times \frac{1}{2} \ln(2)$$

$$= RC \left(1 + \frac{1}{n}\right) \frac{\ln(2)}{2}$$

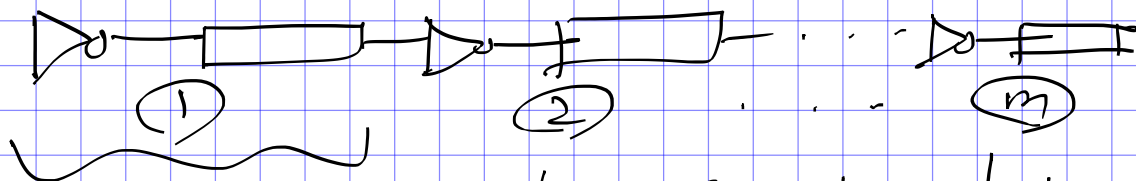
$$\text{for } n \rightarrow \infty \quad \tau = \frac{RC \ln(2)}{2} \quad \left\{ \begin{array}{l} \text{Distributed} \\ \text{RC-line} \end{array} \right.$$

Let $R' = R/L$ and $C' = C/L$

$$\tau = \frac{R' C' L^2 \ln(2)}{2}$$

for every layer PDK provides R', C' .

Let us section the long interconnect into "m" parts and add inverters.



Ignoring inverter delay $\tau_1 = \frac{R' C' L^2 \ln(2)}{2m^2}$ | cap of section $= C' \times \frac{L}{m}$
res. of section $= R' \times \frac{L}{m}$

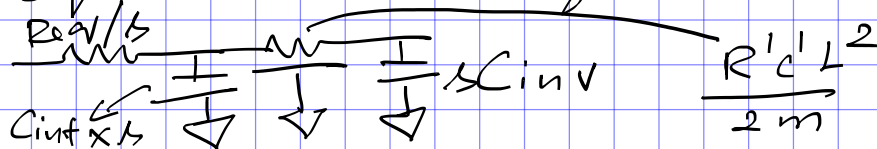
$$\text{Total delay } \tau = m \times \tau_1 = \frac{R' C' L^2}{2m}$$

This indicates that section the long line and adding inverters potentially reduces the delay.

Question then is how many sections & how to size the inverter?

Let s' be the size of each inverter.

The equivalent model of a section is:



$$\tau_{\text{section}} = \frac{R_{eq}}{s} \left(sC_{int} + \frac{C'L}{m} + sC_{inv} \right) \leftarrow \text{inverter delay} \\ + \frac{R^2 C'L^2}{2m^2} \text{ (interconnect)} \\ + \frac{R'L}{m} \times sC_{inv} \text{ (interconnect driving load)}$$

$$\tau_{\text{total}} = m \times \tau_{\text{section}} = R_{eq} (C_{int} \times m + C'L + C_{inv} \times m) \\ + \frac{R^2 C'L^2}{2m} + R'L sC_{inv}$$

To find optimal number of stages

$$\frac{d\tau_{\text{total}}}{dm} = 0 = R_{eq} (C_{int} + C_{inv}) - \frac{R^2 C'L^2}{2m^2}$$

$$m = \sqrt{\frac{R^2 C^2 L^2 \times \frac{1}{4}}{R_{eq} C_{inv}}} = \sqrt{\frac{C_{unbuffered}}{C_{pinv} \times 2}}$$

To find optimal size 's'

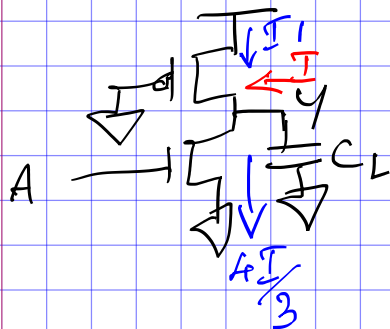
$$\frac{d\tau_{\text{total}}}{ds} = 0 = -\frac{R_{eq} C'L}{s^2} + R'L C_{inv}$$

$$s = \sqrt{\frac{R_{eq} C'L}{R'L C_{inv}}}$$

$$s = \sqrt{\frac{R_{eq} C}{R C_{inv}}}$$

Sizing pseudonMOS to match static CMOS PDN

Assuming pMOS $\left(\frac{W_{min}}{L_{min}}\right)$ provides $\frac{1}{2}$ the current of nMOS $\left(\frac{W_{min}}{L_{min}}\right)$.



For the pseudonMOS to sink the same current from the load C_L as that of a static CMOS

Assume $\left(\frac{W}{L}\right)_n = \frac{4}{3}$

$\Rightarrow$ PMOS has to supply $\left(\frac{4I}{3} - I\right)$ current

$\Rightarrow I_1 = I/3 \Rightarrow \left(\frac{W}{L}\right)_p = \frac{2}{3}$

Calculating 'p' & 'g' of this pseudonMOS
Equivalent inverter

